Notebook-by-Notebook Summary

What each Day 12 notebook covers, why it matters and what it produced

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Notebook-by-notebook overview

Each notebook follows the same print-ready structure: detailed HTML theory, short commented code cells, executed outputs, green output-aware conclusions, no personal paths and no inline installations.

Notebook	Main focus	Datasets	Core models
01 Linear Regression Advertising	Simple and multiple linear regression, OLS, residual diagnostics, statsmodels OLS.	Advertising	Linear Regression, OLS
02 Regularized Regression	Ridge, Lasso, ElasticNet, coefficient shrinkage, Lasso path, LassoCV.	Diabetes regression dataset	Ridge, Lasso, ElasticNet, LassoCV
03 GLM Count Models	Bike Sharing count prediction, Poisson, Negative Binomial, overdispersion.	Bike Sharing	Linear Regression, PoissonRegressor, Poisson GLM, Negative Binomial GLM
04 Logistic Regression	Binary classification, probability, thresholding, calibration and lift.	Customer campaign response	Logistic Regression
05 Advanced Diagnostics	Missing advanced topics and influence diagnostics.	Advertising, Bike Sharing, campaign response	OLS diagnostics, gradient descent, multinomial logistic regression

01 Linear Regression Advertising

Focus: Simple and multiple linear regression, OLS, residual diagnostics, statsmodels OLS.

Dataset family: Advertising

Models and methods: Linear Regression, OLS

Result	Value
TV-only R2	0.6401
TV-only RMSE	3.0064
Multiple regression R2	0.8870
Multiple regression RMSE	1.6846

02 Regularized Regression

Focus: Ridge, Lasso, ElasticNet, coefficient shrinkage, Lasso path, LassoCV.

Dataset family: Diabetes regression dataset

Models and methods: Ridge, Lasso, ElasticNet, LassoCV

Result	Value
Best selected LassoCV alpha	1.6552
LassoCV Test RMSE	52.9196
LassoCV Test R2	0.4714
Zero coefficients	3

03 GLM Count Models

Focus: Bike Sharing count prediction, Poisson, Negative Binomial, overdispersion.

Dataset family: Bike Sharing

Models and methods: Linear Regression, PoissonRegressor, Poisson GLM, Negative Binomial GLM

Result	Value
Best MAE model	Linear Regression
Linear Regression MAE	8.194
Best count deviance model	Negative Binomial GLM
Negative Binomial Mean Poisson Deviance	4.213

04 Logistic Regression

Focus: Binary classification, probability, thresholding, calibration and lift.

Dataset family: Customer campaign response

Models and methods: Logistic Regression

Result	Value
Accuracy	0.777
Precision	0.625
Recall	0.055
ROC-AUC	0.614
PR-AUC	0.349

05 Advanced Diagnostics

Focus: Missing advanced topics and influence diagnostics.

Dataset family: Advertising, Bike Sharing, campaign response

Models and methods: OLS diagnostics, gradient descent, multinomial logistic regression

Topic closed	Evidence
Closed-form OLS	Matches statsmodels coefficients exactly
Gradient descent	Learning curve included
Cook distance/leverage	Influence diagnostics chart
Multinomial logistic regression	Test accuracy 0.700

References and source foundation

The documents use the completed Day 12 notebooks as the main project evidence. The theory explanations are also aligned with the uploaded course materials and official library documentation.

Source	How it was used
Uploaded course guides: leit03.pdf and Haupt_leitfaden03.pdf	OLS, least squares, residuals, regression line, correlation, interpolation/extrapolation.
Uploaded ML slides: Methode der Kleinsten Quadrate and Typen des Machinellen Lernens	Supervised learning, regression vs classification, gradient descent, logistic regression context.
scikit-learn Linear Models User Guide - https://scikit-learn.org/stable/modules/linear_model.html	Linear Regression, Ridge, Lasso, ElasticNet, Logistic Regression, PoissonRegressor.
scikit-learn Model Evaluation User Guide - https://scikit-learn.org/stable/modules/model_evaluation.html	MSE, MAE, R2, ROC-AUC, PR-AUC, precision, recall, F1, calibration.
statsmodels GLM documentation - https://www.statsmodels.org/stable/glm.html	Generalized Linear Models, Poisson family and model interpretation.
statsmodels NegativeBinomial documentation - https://www.statsmodels.org/stable/generated/statsmodels.discrete.discrete_model.NegativeBinomial.html	Negative Binomial implementation details and count-model context.
UCLA IDRE Negative Binomial Regression notes - https://stats.oarc.ucla.edu/r/dae/negative-binomial-regression/	Practical explanation of Negative Binomial as a model for overdispersed count outcomes.